

A GAMMA-DISTRIBUTED STOCHASTIC FRONTIER MODEL

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We modify the stochastic frontier model of Aigner, Lovell, and Schmidt to allow the one-sided part of the disturbance to have a two-parameter Gamma distribution rather than the less flexible half-normal distribution. Maximum-likelihood estimation and the estimation of firm-specific inefficiency estimates require the evaluation of integrals which have no closed form and for which there are no polynomial approximations available. We consider two methods of computing these integrals. We also present a corrected OLS estimator based on the methods of moments. An application is presented for illustration. We find that for these data, the gamma distribution produces results which differ noticeably from those of three alternative formulations.

1. Introduction

A large part of the recent empirical literature on efficiency in production has been carried out in the framework of Aigner, Lovell, and Schmidt's (1977) [and Meeusen and van den Broeck's (1977)] composed error or stochastic frontier model. But, the one-sided disturbance as a model of inefficiency continues to have an appeal. See, for example, Aguilar (1988), Greene (1980), Kopp and Diewert (1982), and Fare, Grosskopf, and Lovell (1985). The half-normal distribution proposed by Aigner, Lovell, and Schmidt is a bit inflexible in that it is a single-parameter distribution and it does embody the assumption that the density of the disturbances is most concentrated near zero. In view of this, Stevenson (1980) suggested shifting the half-normal distribution by allowing a nonzero mode, producing a general truncated normal distribution instead. Stevenson also proposed a restricted version of the Gamma model analyzed here. Thus, various methods of relaxing the restrictions implicit in the half-normal have appeared in the literature. Still, the model of Aigner, Lovell, and Schmidt continues to dominate the received empirical analysis.

This paper will report development of an alternative model based on the Gamma distribution instead of the half-normal. The Gamma frontier model was proposed by Greene (1980) in the context of the deterministic frontier model:

$$y = f(x, \beta) - u \quad \text{where} \quad u \sim G(\theta, P),$$

so

$$f(u) = \frac{\Theta^P}{\Gamma(P)} u^{P-1} e^{-\Theta u}, \quad u \geq 0, \quad \Theta, P > 0. \quad (1)$$

[The preceding and all of the derivations to follow employ the notation of Aigner, Lovell, and Schmidt (1977) in order to facilitate comparison of the results.] Maximum-likelihood estimation and corrected least-squares estimation are discussed in Greene (1980), to which the reader is referred for further details. The salient features of this model are, foremost, its one-sided disturbance, and, second, the useful features of the Gamma distribution. In particular, the asymmetry of the distribution is contained in the parameter P ; Θ is just a scale parameter. The skewness coefficient is larger the smaller is P . The efficiency gain of maximum-likelihood estimation over least-squares estimation in this model is $P/(P-2)$. The distribution approaches symmetry as P goes to infinity. The OLS constant term remains inconsistent, however.¹

In spite of their aesthetic appeal, one-sided disturbances have a serious practical shortcoming. Any measurement error in the dependent variable will have extremely detrimental effects on the analysis, and a single errant observation can dominate even a large sample. We propose in this paper to combine the Gamma frontier model with Aigner et al.'s stochastic frontier specification. Thus, the model to be analyzed here is

$$y = f(\mathbf{x}, \boldsymbol{\beta}) + v - u,$$

where

$$v \sim N[0, \sigma^2] \quad \text{and} \quad u \sim G[\Theta, P]. \quad (2)$$

This refinement of the stochastic frontier model was suggested by Stevenson (1980), who gives results for a few integer values of P . A formulation of the log-likelihood and its gradient appears in Beckers and Hammond (1987),² but no practical application of the model has appeared previously.

The plan of this paper is as follows: Section 2 will briefly review the formulation of the stochastic frontier model to provide more detail on the specific context of this model. In section 3, we derive a formulation of the probability distribution of the difference between a normally distributed random variable and one with a Gamma distribution. In section 4, we present a maximum-likelihood estimator and procedures for evaluating these integral expressions. This section will also present an estimator of the firm-specific inefficiency component. In section 5, we present a corrected ordinary least-

¹For analysis of this point, see Deprins and Simar (1986).

²They present alternative forms of our (29), (33), (35), (37), and (42).

squares estimator that could be used in its own right, though we are primarily interested in a consistent starting value for the maximum-likelihood estimator. Some empirical estimates are given in section 6 to illustrate the technique. Conclusions are drawn in section 7.

2. The stochastic frontier model

The stochastic frontier model departs from an idealized production function,

$$y^* = f(\mathbf{x}, \boldsymbol{\beta}). \quad (3)$$

For any producer, the frontier is stochastic for a variety of reasons including, for example, simple good luck. Thus, for any individual firm,

$$y^\circ = f(\mathbf{x}, \boldsymbol{\beta}) + v. \quad (4)$$

Inefficiency enters the production model through a positive disturbance, u , that is independent of v . For the i th firm,

$$y_i = y_i^\circ - u_i = f(\mathbf{x}_i, \boldsymbol{\beta}) + v_i - u_i = f(\mathbf{x}_i, \boldsymbol{\beta}) + \varepsilon_i. \quad (5)$$

Aigner, Lovell, and Schmidt present the probability distribution of ε under the assumption that v is distributed as $N[0, \sigma_v^2]$ and u is the absolute value of a variable which is independent of v and is distributed as $N[0, \sigma_u^2]$. The log-likelihood for a sample of N observations is

$$\ln L = -(N/2)(\ln 2\pi + \ln \sigma^2) + \sum_i \left[\ln \Phi[-\varepsilon_i \lambda / \sigma] - \frac{1}{2}(\varepsilon_i / \sigma)^2 \right], \quad (6)$$

where $\Phi[\cdot]$ and $\phi[\cdot]$ (used below) are the CDF and PDF of the standard normal distribution, respectively, $\sigma^2 = \sigma_v^2 + \sigma_u^2$ and $\lambda = \sigma_u / \sigma_v$.

Once the parameters are estimated, primary interest centers on the estimation of inefficiency, u_i . $E[u_i]$ is a summary measure which, for obvious reasons, is not particularly satisfactory. Jondrow, Lovell, Materov, and Schmidt (1982) suggest a firm-specific estimate,

$$E[u_i | v_i - u_i] = E[u_i | \varepsilon_i] = \frac{\sigma \lambda}{1 + \lambda^2} \left[\frac{\phi[\varepsilon_i \lambda / \sigma]}{\Phi[\varepsilon_i \lambda / \sigma]} - \frac{\varepsilon_i \lambda}{\sigma} \right]. \quad (7)$$

For our purposes in this study, Stevenson's (1980) extension of the model is important. He analyzes the case in which u , still truncated at zero, has a truncated normal distribution with parameters μ , which is allowed to differ from zero in either direction, and σ_u^2 . Details on this model, including the applicable log-likelihood function used in estimation, may be found in his paper. The counterpart to (7) (which he does not present) is obtained by replacing $\varepsilon_i\lambda/\sigma$ in (7) with

$$\frac{\alpha}{\sigma^*} = \frac{\sigma_v^2\mu + \sigma_u^2\varepsilon_i}{\sigma_v^2 + \sigma_u^2} \left[\frac{\sigma_v^2\sigma_u^2}{\sigma_v^2 + \sigma_u^2} \right]^{-1/2} = \frac{\varepsilon_i\lambda}{\sigma} + \frac{\mu}{\sigma\lambda}.$$

3. Combining the gamma distribution with the stochastic frontier

We now consider a stochastic frontier model, (2), in which u_i has the gamma distribution in (1). This specification enjoys essentially the same properties as normal/half-normal model with the additional advantage of the flexibility of a two-parameter distribution. The primary advantage is that it does not require that the firm-specific inefficiency measures be predominately near zero. For values of P between 0 and 1, it has the shape of the exponential (which has also been proposed as an alternative model,³ though it has rarely been used empirically). Thus, for these values, the mass of the distribution is still concentrated near zero. But, for values of P greater than 1, the terminal value of $f(u)$ is zero. As such, the distribution of disturbances is concentrated at a point away from zero. The larger is P , the greater is this effect. (The mean is P/Θ .) Note that in Greene's (1980) paper, the requirement that P be greater than 2 implied this second general characteristic. This restriction was needed to ensure regularity of the log-likelihood function. Since the range of the random variable, ε , is no longer restricted, there is no such requirement in the model we consider here. We only require that P be positive. The distribution of inefficiency can have many different shapes.

The density of a gamma distributed variable is given in (1). The expected value and variance of the disturbance in the frontier model (2) are

$$E[\varepsilon] = E[v - u] = -E[u] = -P/\Theta \quad (8)$$

and

$$\text{var}[\varepsilon] = \text{var}[v] + \text{var}[u] = \sigma_v^2 + P/\Theta^2. \quad (9)$$

The third central moment for a gamma-distributed variate can be derived by

³See Meeusen and van den Broeck (1977).

expanding the raw moment and using the result for the gamma distribution,

$$E[u^r] = (\Gamma(P+r)/\Gamma(P))/\Theta^r. \quad (10)$$

Thus, the skewness coefficient for this disturbance is

$$E[(u - E[u])^3] = 2P/\Theta^3. \quad (11)$$

For purposes of estimation, we require the density of the compound disturbance $\varepsilon = v - u$. The joint density of ε and u is

$$f(\varepsilon, u) = f_v(\varepsilon + u)f_u(u). \quad (12)$$

Now, insert the normal density of v and the gamma density of u to obtain

$$f(\varepsilon, u) = (2\pi\sigma^2)^{-1/2} e^{-1/2(\varepsilon+u)^2/\sigma^2} \frac{\Theta^P}{\Gamma(P)} u^{P-1} e^{-\Theta u}. \quad (13)$$

We now expand this and collect terms. Thus,

$$f(\varepsilon, u) = (2\pi\sigma^2)^{-1/2} e^{-\varepsilon^2/2\sigma^2} e^{-u^2/2\sigma^2} e^{-\varepsilon u/\sigma^2} \frac{\Theta^P}{\Gamma(P)} u^{P-1} e^{-\Theta u}. \quad (14)$$

Gathering terms in u ,

$$\begin{aligned} f(\varepsilon, u) &= \frac{(2\pi\sigma^2)^{-1/2} \Theta^P}{\Gamma(P)} e^{-\varepsilon^2/2\sigma^2} e^{-u^2/2\sigma^2 - \varepsilon u/\sigma^2 - \Theta u} u^{P-1} \\ &= K e^{-\varepsilon^2/2\sigma^2} e^{-u^2/2\sigma^2 - u(\varepsilon/\sigma^2 + \Theta)} u^{P-1}, \end{aligned} \quad (15)$$

where K is the leading product of constants. Then,

$$f(\varepsilon, u) = K e^{-\varepsilon^2/2\sigma^2} e^{-1/2[u^2 + 2u\sigma^2(\varepsilon/\sigma^2 + \Theta)]/\sigma^2} u^{P-1}. \quad (16)$$

Completing the square in the second exponent produces

$$\begin{aligned} f(\varepsilon, u) &= K e^{-\varepsilon^2/2\sigma^2} e^{-1/2[u^2 + 2u(\varepsilon + \Theta\sigma^2) + (\varepsilon + \Theta\sigma^2)^2 - (\varepsilon + \Theta\sigma^2)^2]/\sigma^2} u^{P-1} \\ &= K e^{-\varepsilon^2/2\sigma^2 + 1/2(\varepsilon + \Theta\sigma^2)^2/\sigma^2} e^{-1/2[u + (\varepsilon + \Theta\sigma^2)]^2/\sigma^2} u^{P-1}. \end{aligned} \quad (17)$$

To obtain the marginal distribution of ε , we must integrate u out of this expression. The first term does not involve u . Carrying part of K inside the integral produces

$$h = \int_0^\infty (2\pi\sigma^2)^{-1/2} e^{-1/2[u + (\varepsilon + \Theta\sigma^2)]^2/\sigma^2} u^{P-1} du. \quad (18)$$

Let $Q|\varepsilon$ be a normally distributed variable with mean $-(\varepsilon + \Theta\sigma^2)$ and variance σ^2 . Then,

$$\int_0^\infty (2\pi\sigma^2)^{-1/2} e^{-1/2[Q + (\varepsilon + \Theta\sigma^2)]^2/\sigma^2} dQ = \text{Prob}[Q > 0|\varepsilon]. \quad (19)$$

As such, after cancelling the constants, we find that

$$\frac{e^{-1/2[Q + (\varepsilon + \Theta\sigma^2)]^2/\sigma^2}}{\int_0^\infty e^{-1/2[Q + (\varepsilon + \Theta\sigma^2)]^2/\sigma^2} dQ} = f(Q|Q > 0, \varepsilon). \quad (20)$$

This is the density of a random variable with truncated normal distribution. Therefore, the desired integral in (18) can be written as

$$\begin{aligned} & \text{Prob}[Q > 0|\varepsilon] \int_0^\infty Q^{P-1} f(Q|Q > 0, \varepsilon) dQ \\ &= \text{Prob}[Q > 0|\varepsilon] E[Q^{P-1}|Q > 0, \varepsilon], \end{aligned} \quad (21)$$

where Q has the truncated normal distribution in (20).

Returning to the marginal density, we have

$$\begin{aligned} f(\varepsilon) &= \frac{\Theta^P}{\Gamma(P)} e^{-1/2\varepsilon^2/\sigma^2 + 1/2(\varepsilon + \Theta\sigma^2)^2/\sigma^2} \\ &\quad \times \text{Prob}[Q > 0|\varepsilon] E[Q^{P-1}|Q > 0, \varepsilon]. \end{aligned} \quad (22)$$

Expanding the exponent and collecting terms, we arrive, finally, at

$$f(\varepsilon) = \frac{\Theta^P}{\Gamma(P)} e^{\Theta\varepsilon + \sigma^2\Theta^2/2} \text{Prob}[Q > 0|\varepsilon] E[Q^{P-1}|Q > 0, \varepsilon], \quad (23)$$

where $Q|\varepsilon$ is normally distributed with mean and variance

$$E[Q|\varepsilon] = -(\varepsilon + \Theta\sigma^2) \quad \text{and} \quad \text{var}[Q|\varepsilon] = \sigma^2. \quad (24)$$

If

$$h(r, \varepsilon) = E[Q^r | Q > 0, \varepsilon], \quad (25)$$

then

$$f(\varepsilon) = \frac{\Theta^P}{\Gamma(P)} e^{\Theta\varepsilon + \sigma^2\Theta^2/2} \text{Prob}[Q > 0|\varepsilon] h(P-1, \varepsilon). \quad (26)$$

The latter term in $f(\varepsilon)$ is a fractional moment of the truncated normal distribution. Since P need not be an integer, there will be no closed form for $h(P-1, \varepsilon)$ and thus none for the density.

Finally, for estimation of a cost frontier rather than a production frontier, where u enters positively rather than negatively in (5), we must change the sign of ε in the exponent and the mean of Q to $\varepsilon - \Theta\sigma^2$.

We are also interested in estimation of the conditional moments of the efficiency component, u . Following Jondrow et al., we obtain the conditional mean function, $E[u|\varepsilon]$. To derive $E[u|\varepsilon]$, we begin with

$$f(u|\varepsilon) = f(\varepsilon, u)/f(\varepsilon).$$

Inserting our earlier results and cancelling terms in the fraction, we find

$$f(u|\varepsilon) = \frac{(2\pi\sigma^2)^{-1/2} e^{-1/2[u+(\varepsilon+\Theta\sigma^2)]^2/\sigma^2} u^{P-1}}{\text{Prob}[Q > 0|\varepsilon] h(P-1, \varepsilon)}. \quad (27)$$

Jondrow et al. present the counterpart to (27) for the special case in which u has an exponential distribution. They find the conditional distribution is a truncated normal distribution where the untruncated structural normal variable has mean $-(\varepsilon + \Theta\sigma^2)$ and variance σ^2 (using our notation). Their result follows after the substitution $\sigma = \sigma_u$ and $\Theta = 1/\sigma_u$. The exponential case corresponds to $P = 1$. If $P = 1$, in (27), $h(P-1, \varepsilon) = 1$ and the truncated normal distribution emerges.

Multiplying and dividing (27) by $\text{Prob}[Q > 0|\varepsilon]$ produces

$$E[u|\varepsilon] = \int_0^\infty u f(u|\varepsilon) du = h(P, \varepsilon)/h(P-1, \varepsilon). \quad (28)$$

Jondrow et al. (1982, (5) on p. 236) give an expression for $E[u|\varepsilon]$ for the case in which u has an exponential distribution. With $P = 1$, $h(P - 1, \varepsilon)$ equals 1 while $h(P, \varepsilon)$ is just the expected value $E[Q|Q > 0]$ when Q has mean $-(\varepsilon + \Theta\sigma^2)$. After this substitution, as expected, their (5) emerges as the special case.⁴

4. Maximum-likelihood estimation

The log-likelihood function for a sample of N observations is

$$\begin{aligned}\ln L &= \sum_i \ln f[\varepsilon_i] \\ &= \sum_i \left[P \ln \Theta - \ln \Gamma(P) + \Theta^2 \sigma^2 / 2 + \Theta \varepsilon_i \right. \\ &\quad \left. + \ln \text{Prob}[Q > 0 | \varepsilon_i] + \ln h[P - 1, \varepsilon_i] \right].\end{aligned}\quad (29)$$

The log-likelihood cannot be evaluated in the form given above. The standard normal integral, $\text{Prob}[Q > 0 | \varepsilon] = \Phi[-(\varepsilon + \Theta\sigma^2)/\sigma]$ can be approximated by the familiar polynomial approximation. The one given by Abramovitz and Stegun (1964) is typically used. But this leaves (18), the integral in the expectation function, $h(P - 1, \varepsilon)$.

For purposes of estimation, we used a 15-point Laguerre quadrature.⁵ This was fast and accurate and replicated the computation of the exponential model (see section 6 below) in which computation of $h(r, \varepsilon)$ is bypassed, to within a difference of less than 1%. But, we found that in computation of $E[u|\varepsilon]$, the quadrature formula performed rather poorly in the tails of the distribution (when $E[u|\varepsilon]$ was very small), so in order to calculate the ratio $h(P, \varepsilon)/h(P - 1, \varepsilon)$ at the MLE, we used instead a Newton-Cotes quadrature with 500 segments.^{6, 7} We required two modifications of this rule for our purposes. The range of integration in (18) is $[0, \infty)$, so the trapezoid rule is not quite appropriate. But, the normal distribution has finite moments of all orders. Thus, in the tails of the distribution, $f(u)$ must fall to zero. As an approximation, we applied the rule over the range 0 to +5 standard devia-

⁴The same procedure produces $E[u^2 | \varepsilon] = h(P + 1, \varepsilon)/h(P - 1, \varepsilon)$ which can be used to find the variance. For integer powers, $E[u^r | \varepsilon] = h(P + r - 1, \varepsilon)/h(P - 1, \varepsilon)$.

⁵See Abramovitz and Stegun (1964, p. 890).

⁶For discussion, see Kennedy and Gentle (1980, pp. 87-88).

⁷The Monte Carlo method of integration [see, e.g., McFadden (1989) or Manski and Lerman (1977)] is another possibility. However, we found the sample sizes needed to obtain stable enough values of the function to use in an iterative procedure were too large (over 5000) to be practical.

tions of the range of Q . If P is less than 1, the terminal value is not computable when we evaluate $h(P-1, \varepsilon)$. To avoid the division by zero, we used +0.0001 standard deviations instead of zero for a . For our applications we used 500 segments. For a few known values, this was accurate to six digits. Although far too slow for estimation, this method was preferable for computing the MLE of $E[u|\varepsilon]$.

We now obtain the derivatives of the log-likelihood. Write

$$\begin{aligned} \ln f(\varepsilon) &= P \ln \Theta - \ln \Gamma(P) + \sigma^2 \Theta^2 / 2 + \Theta \varepsilon \\ &\quad + \ln \int_0^\infty (2\pi\sigma^2)^{-1/2} e^{-1/2[u+\varepsilon+\Theta\sigma^2]^2/\sigma^2} u^{P-1} du. \end{aligned} \quad (30)$$

Since the limits of integration are not dependent on the parameters, we may pass the differentiation through the integral. Thus,

$$\begin{aligned} \frac{\partial \ln f}{\partial \varepsilon} &= \Theta + \frac{\int_0^\infty (2\pi\sigma^2)^{-1/2} [-(u+\varepsilon+\Theta\sigma^2)/\sigma^2] e^{-1/2[u+\varepsilon+\Theta\sigma^2]^2/\sigma^2} u^{P-1} du}{\int_0^\infty (2\pi\sigma^2)^{-1/2} e^{-1/2[u+\varepsilon+\Theta\sigma^2]^2/\sigma^2} u^{P-1} du} \\ &\quad (31) \end{aligned}$$

The ratio of integrals is a sum of three terms. But, the second and third do not involve u , so they may be factored out of the ratio, which then evaluates to one. The third term, involving $-u$, is of the same form we used to obtain $E[u|\varepsilon]$. Combining terms and using the results derived earlier, we find

$$\begin{aligned} \frac{\partial \ln f}{\partial \varepsilon} &= -(1/\sigma^2) \left[\varepsilon + \frac{h(P, \varepsilon)}{h(P-1, \varepsilon)} \right] \\ &= -(1/\sigma^2)(\varepsilon + E[u|\varepsilon]). \end{aligned} \quad (32)$$

From the original specification, $\varepsilon = y - \beta'x$, so

$$\frac{\partial \ln f}{\partial \beta} = (1/\sigma^2) \left[\varepsilon + \frac{h(P, \varepsilon)}{h(P-1, \varepsilon)} \right] x. \quad (33)$$

The derivatives of the log-likelihood are obtained by summing this expression over the N observations.

Proceeding in similar fashion,

$$\frac{\partial \ln f}{\partial \Theta} = \frac{P}{\Theta} + \varepsilon + \Theta \sigma^2 + \frac{\int_0^\infty (2\pi\sigma^2)^{-1/2} [-(u + \varepsilon + \Theta\sigma^2)] e^{-1/2[u + \varepsilon + \Theta\sigma^2]^2/\sigma^2} u^{P-1} du}{\int_0^\infty (2\pi\sigma^2)^{-1/2} e^{-1/2[u + \varepsilon + \Theta\sigma^2]^2/\sigma^2} u^{P-1} du} \quad (34)$$

As before the latter two of the three terms in the integral become constants times one, and the third reduces as previously, so we obtain

$$\frac{\partial \ln f}{\partial \Theta} = \frac{P}{\Theta} - \frac{h(P, \varepsilon)}{h(P-1, \varepsilon)} = E[u] - E[u|\varepsilon]. \quad (35)$$

Continuing,

$$\frac{\partial \ln f}{\partial P} = \ln \Theta - \frac{\Gamma'(P)}{\Gamma(P)} + \frac{\int_0^\infty (2\pi\sigma^2)^{-1/2} \ln u e^{-1/2[u + \varepsilon + \Theta\sigma^2]^2/\sigma^2} u^{P-1} du}{\int_0^\infty (2\pi\sigma^2)^{-1/2} e^{-1/2[u + \varepsilon + \Theta\sigma^2]^2/\sigma^2} u^{P-1} du} \quad (36)$$

This has the same form as several earlier expressions. Dividing numerator and denominator by the relevant probability produces

$$\frac{\partial \ln f}{\partial P} = \ln \Theta - \frac{\Gamma'(P)}{\Gamma(P)} + \frac{E[Q^{P-1} \ln Q | Q > 0, \varepsilon]}{h(P-1, \varepsilon)}. \quad (37)$$

The latter part of (37) may be recognized as $E[\ln u|\varepsilon]$, which produces a useful formulation of $\partial \ln f/\partial P$,

$$\frac{\partial \ln f}{\partial P} = \ln \Theta - \Gamma'(P)/\Gamma(P) + E[\ln u|\varepsilon]. \quad (38)$$

To differentiate with respect to σ^2 , it is convenient to define

$$q = (2\pi\sigma^2)^{-1/2} e^{-1/2[u + \varepsilon + \Theta\sigma^2]^2/\sigma^2} u^{P-1}. \quad (39)$$

This is the weighting function in the preceding integrals. We will require

$$\frac{\partial \ln f}{\partial \sigma^2} = \Theta^2/2 + \frac{\int_0^\infty \partial q / \partial \sigma^2 du}{\int_0^\infty q du}. \quad (40)$$

Let $w = u + \varepsilon + \Theta \sigma^2$. Then,

$$\frac{\partial q}{\partial \sigma^2} = \frac{-q}{2\sigma^2} + \frac{q}{\sigma^2} \left[\frac{w^2}{2\sigma^2} - \Theta w \right].$$

After some tedious algebra, this reduces to

$$\frac{\partial q}{\partial \sigma^2} = -\frac{q}{\sigma^2} \left[\frac{1}{2} - \frac{\varepsilon^2}{2\sigma^2} + \frac{\Theta^2 \sigma^2}{2} + \frac{u^2}{2\sigma^2} - \frac{\varepsilon u}{\sigma^2} \right]. \quad (41)$$

In $\partial \ln f / \partial \sigma^2$, the terms in $\partial q / \partial \sigma^2$ which do not involve u can be moved outside the integral, and the ratio of integrals will be one. The terms involving u and u^2 are of the form derived earlier as ratios of the function $h(r, \varepsilon)$. The third term in brackets above integrates simply to $-\Theta^2/2$ which cancels the leading term in the derivative. Combining all of these results produces

$$\frac{\partial \ln f}{\partial \sigma^2} = \frac{1}{2\sigma^2} \left[\frac{\varepsilon^2}{\sigma^2} - 1 \right] + \frac{h(P+1, \varepsilon) + 2\varepsilon h(P, \varepsilon)}{2(\sigma^2)^2 h(P-1, \varepsilon)}. \quad (42)$$

As in (33), the effect of the asymmetry is to displace the familiar result for the classical normal regression model. As before, this can be written as the classical regression counterpart plus a moment of $u|\varepsilon$:

$$\frac{\partial \ln f}{\partial \sigma^2} = \frac{1}{2\sigma^2} \left[\frac{\varepsilon^2}{\sigma^2} - 1 \right] + \frac{1}{2(\sigma^2)^2} [E[u^2|\varepsilon] + 2\varepsilon E[u|\varepsilon]]. \quad (43)$$

The derivatives are useful to improve the search for a maximum-likelihood estimator, since the numerical derivatives may have some rounding error. This is likely to be more problematic than usual in the current context because of the need to approximate $h(r, \varepsilon)$. They will also be usable for estimating the asymptotic covariance matrix of the MLE using the Berndt et al. (1974) estimator.

5. A consistent method-of-moments estimator based on OLS

As has been documented a number of times in the received literature, in the stochastic frontier model (and the one-sided gamma frontier model as well), the OLS slope estimators are consistent, but the OLS constant term is biased. It is offset by the nonzero mean of the disturbance. Greene (1980) obtained estimators for the parameters of the disturbance distribution, and thus the constant term, in the gamma frontier model by manipulating the OLS residuals. [Waldman (1982) presents a counterpart for the normal/half-normal model. Gabrielsen (1975) and Aguilar (1988) present some general results on the method of proof used in Greene (1980).] A similar procedure can be used here. Note, first,

$$E[\varepsilon] = E[v - u] = E[v] - E[u] = -E[u] = -P/\Theta.$$

The least-squares residuals have mean zero by construction, so the mean residual is not useful. But, higher central moments of ε can be consistently estimated by using the counterpart for the OLS residuals. Thus,

$$s^2 = e'e/N$$

is a consistent estimator of

$$E[(\varepsilon - E[\varepsilon])^2] = \text{var}[\varepsilon] = \text{var}[u] + \text{var}[v] = \sigma^2 + P/\Theta^2.$$

In obtaining the third and fourth moments, it is useful to note that

$$E[(\varepsilon - E[\varepsilon])^r] = E[(v - (u - E[u]))^r].$$

Since $E[v] = 0$, we can easily expand these functions for values of r using the binomial expansions, and, thereafter, use the well-known central moments for the normal and gamma distributions. Thus, taking the third and fourth powers, we have

$$\begin{aligned} E[(\varepsilon - E[\varepsilon])^3] &= E[v^3] - 3E[v^2]E[u - E[u]] \\ &\quad + 3E[v]E[(u - E[u])^2] - E[(u - E[u])^3]. \end{aligned}$$

All terms are zero except the last. The third central moment of the gamma

distribution is

$$E[(u - E[u])^3] = 2P/\Theta^3,$$

so it follows that

$$\text{plim}(1/N) \sum_i e_i^3 = -2P/\Theta^3.$$

Using the same device, we find that

$$E[(\varepsilon - E[\varepsilon])^4] = 3\sigma^4 + 6\sigma^2 P/\Theta^2 + 3P(P+2)/\Theta^4.$$

For purposes of manipulating the moments, it is simpler to work with

$$E[(\varepsilon - E[\varepsilon])^4] - 3(\text{var}[\varepsilon])^2 = 6P/\Theta^4.$$

This moment would be zero if the disturbance were normally distributed. Thus, it may be viewed as reflecting the departure from normality. Using the standard notation, we have

$$\begin{aligned} \text{plim}(1/N) \sum_i e_i^2 &= \text{plim } m_2 = \sigma^2 + P/\Theta^2, \\ \text{plim}(1/N) \sum_i e_i^3 &= \text{plim } m_3 = -2P/\Theta, \\ \text{plim}(m_4 - 3m_2^2) &= \text{plim } m_4^* = 6P/\Theta^4. \end{aligned} \tag{44}$$

The method of moments estimators are

$$\begin{aligned} \hat{\Theta} &= -3m_3/m_4^*, \\ \hat{P} &= -\hat{\Theta}^3 m_3/2, \\ \hat{\sigma}^2 &= m_2 - \hat{P}/\hat{\Theta}^2. \end{aligned} \tag{45}$$

Only the first two are actually needed to correct the OLS intercept, though the variance is also useful for other purposes. Note that the procedure breaks down if the third moment of the residuals is positive while Θ and P will both be zero if the residuals are distributed symmetrically.⁸

⁸In this connection, see the results of Waldman (1982).

This gives a complete set of estimates. The OLS estimated standard errors are appropriate for all coefficients save for the intercept. The appropriate asymptotic standard error for the corrected constant term and for the estimates of the nuisance parameters can be obtained using Newey's (1984) results for GMM estimators. Since these are generally not needed for inference in this model, in the interest of brevity, they are omitted. In principle, we can also use the earlier results to compute conditional estimates of the firm-specific efficiency values, $E[u|\varepsilon]$. The corrected OLS estimates can also be used as starting for an iterative maximum-likelihood procedure or to begin the two-step procedure described below.

Since expressions for the first derivatives have been obtained, a single Newton-like step from these initial consistent estimates using the BHHH asymptotic covariance estimator provides a set of estimates which have the same asymptotic properties as the MLE's.

6. Application

To illustrate the preceding, we have used the data from Christensen and Greene's (1976) study of the U.S. electric utility industry. The model to be fit is a cost function rather than a production function.⁹ We use one of the restricted specifications for the cost function,

$$\begin{aligned} \ln(\text{Cost}/P_f) = & \beta_0 + \beta_1 \ln Q + \beta_2 \ln^2 Q + \beta_3 \ln(P_l/P_f) \\ & + \beta_4 \ln(P_k/P_f) + \varepsilon. \end{aligned} \quad (46)$$

Output (Q) is a function of three factors, labor (l), capital (k), and fuel (f). The three factor prices are P_l , P_k , and P_f . The restriction of linear homogeneity in the factor prices has been imposed *a priori* in the cost function. Further details on the model and data construction can be found in Christensen and Greene. The data used to estimate the model appear in the appendix. Since we are estimating a cost frontier rather than a production frontier, the disturbance in (2) is $v + u$ rather than $v - u$. Some minor changes are required in several results and derivations. For the convenience of the interested practitioner, those in the estimating equations are as

⁹See Schmidt and Lovell (1979) for discussion of the two formulations.

¹⁰Without the factor share equations, estimation of Christensen and Greene's full translog model in this framework was hampered by a severe problem of multicollinearity. The correct formulation of the relationship between the disturbances in the share equations and the efficiency component in the cost function remains to be derived. Thus, the estimates presented here are intended only to be illustrative of the differences which arise when the several models are applied to the same data.

follows:

- (1) In $f(\varepsilon)$, the exponent $\Theta\varepsilon$ is changed to $-\Theta\varepsilon$ [see (23) and (29)].
- (2) In all results using $h(r, \varepsilon)$, the mean of Q is now $\varepsilon - \Theta\sigma^2$.
- (3) The sign of ε is changed in (34).
- (4) The last term in the brackets in (41) is now added.
- (5) The terms $2\varepsilon h(P, \varepsilon)$ in (42) and $2\varepsilon E[u|\varepsilon]$ in (43) are now subtracted.

Table 1 lists the Aigner et al. stochastic frontier and the corrected ordinary least-squares estimates of the parameters in (46). The COLS estimates differ from OLS only in the adjusted constant term, which is the OLS constant plus the estimate of P/Θ . We have also computed the estimates for the exponential model which results when P is constrained to equal one in the gamma model. With this restriction, the log-likelihood and derivatives simplify considerably:

$$\ln f(\varepsilon_i) = \ln \Theta + \Theta^2 \sigma^2 / 2 - \Theta \varepsilon_i + \ln \Phi[(\varepsilon_i - \Theta \sigma^2) / \sigma]$$

(with the sign of ε_i reversed for a production frontier). Maximization of the log-likelihood is straightforward and does not require computation of $h(r, \varepsilon)$. For convenience, this is parameterized in terms of σ instead of σ^2 using the formulation suggested in Aigner et al. Here, our Θ equals their $1/\phi$. Finally, we have obtained the estimates of Stevenson's generalization of the model. Table 1 shows the maximum-likelihood estimates for the various formulations for this model. The maximum-likelihood estimates of the gamma frontier model were computed using the estimates from the exponential model as the starting point for the DFP iterations.

The estimates of the parameters of the cost function obtained with the four models are roughly similar. Stevenson's extension of the ALS model is rejected by the Wald statistic.¹¹ It is worth noting, however, the very large changes in the variance parameters which arise with this model. The hypothesis of the exponential model is also strongly rejected by the likelihood ratio statistic of 91.28. Likewise, the Lagrange multiplier statistic is 123.00 which leads to the same conclusion. It is interesting that the Wald statistic is so small (1.73), but the power of the test in the presence of such a different value of Θ may be problematic. The fact that the full translog model is known to be a preferable specification for these data¹² suggests that these specification tests may be picking up some of the effect of the missing variables as well as the differences in the disturbance processes assumed.

¹¹The expressions for $E[u]$ and $\text{var}[u]$ in Stevenson (1980) are in error. The calculations above are based on the moments of the truncated normal distribution.

¹²See Christensen and Greene (1976).

Table 1
Parameter estimates for the stochastic frontier models (estimated asymptotic standard errors are given in parentheses).

	Corrected OLS		ALS		Stevenson		Gamma		Exponential	
β_0	-7.214	(0.752)	-7.408	(0.292)	-7.501	(0.313)	-7.810	(0.378)	-7.525	(0.311)
β_1	0.3858	(0.0383)	0.4078	(0.0293)	0.4292	(0.0381)	0.4728	(0.0427)	0.4343	(0.0390)
β_2	0.0316	(0.00269)	0.03059	(0.00218)	0.02945	(0.00262)	0.02581	(0.00294)	0.02916	(0.00263)
β_3	0.2470	(0.0670)	0.2445	(0.0646)	0.2501	(0.0649)	0.2910	(0.0677)	0.2521	(0.0624)
β_4	0.0784	(0.0617)	0.05872	(0.0686)	0.0422	(0.0648)	0.02527	(0.0665)	0.03911	(0.0639)
λ			1.4496	(0.342)	3.884	(16.2)				
σ			0.1887	(0.0282)	0.4180	(1.751)	0.1352	(0.00876)	0.1047	(0.0139)
σ^2			0.0356	(0.0106)	0.1747	(1.463)	0.01827	(0.00237)	0.01096	(0.00291)
μ			0.0	(fixed)	-1.377	(14.55)				
P	0.0164						2.4500	(1.102)	1.0000	(fixed)
θ	2.529						21.347	(3.944)	10.12	(2.681)
Log-L			66.14		66.911		112.72		67.09	
$E[u]$	0.0065		0.0988		0.0720		0.1146		0.0988	
$\text{var}[u]$	0.00257		0.00875		0.00616		0.00536		0.00976	
$\text{var}[\nu]$	0.0199		0.0115		0.00732		0.0183		0.0110	
$\text{var}[\varepsilon]$	0.0225		0.0202		0.0135		0.0237		0.0207	
$\text{var}[u]/\text{var}[\varepsilon]$	0.1142		0.4320		0.4570		0.2262		0.4710	

Table 2
Estimated efficiency distributions.

	Half-normal	Truncated normal	Gamma models		
			COLS	Gamma	Exponential
Mean	0.1234	0.1039	0.0295	0.1051	0.0989
Std. dev.	0.0652	0.0759	0.0315	0.0293	0.0770
Minimum	0.0304	0.0243	0.0073	0.0001	0.0234
Maximum	0.3917	0.4858	0.2253	0.2055	0.5041

Table 2 summarizes the estimates of $E[u|\varepsilon]$ based on (7) for the ALS model and Stevenson's extension and (27) for the gamma and exponential models. Once again, for the exponential model, the computation is considerably simpler:

$$E[u|\varepsilon] = \gamma + \sigma \frac{\phi[\gamma/\sigma]}{\Phi[\gamma/\sigma]},$$

where γ equals $\varepsilon - \theta\sigma^2$ for a cost frontier and $-(\varepsilon - \theta\sigma^2)$ for a production frontier.¹³ The two distributions of disturbances based on the truncated normal distribution are quite similar to each other, but rather different from those based on the gamma model. Based on the half-normal frontier model we find that $\text{var}[u]$ accounts for 43.2% of the estimated variance of ε . The exponential model is similar; the counterpart is 47.1%. But, the gamma model allocates only $(P/\theta^2)/(\sigma^2 + P/\theta^2) = 22.6\%$ of the total variance of the disturbance to the inefficiency term.

Fig. 1 shows the estimated density functions for the four models. The gamma model is alone among the four in its nonzero mode. It appears that the estimated inefficiencies based on the gamma model are generally smaller than those based on the half-normal. Inspection of the data suggests this as well. It is noteworthy that the simple correlation of the inefficiency estimates based on the half-normal and the COLS/gamma estimates is 0.906, while that of the MLE/gamma estimates with the ALS estimates is 0.918. We find that in those cases in which the ALS model suggests a very large disturbance, the gamma model follows suit, though with a smaller estimate. For those disturbances not obviously in the tail the gamma model predicts values very similar to the other two.

¹³The individual computed estimates are contained in the appendix.

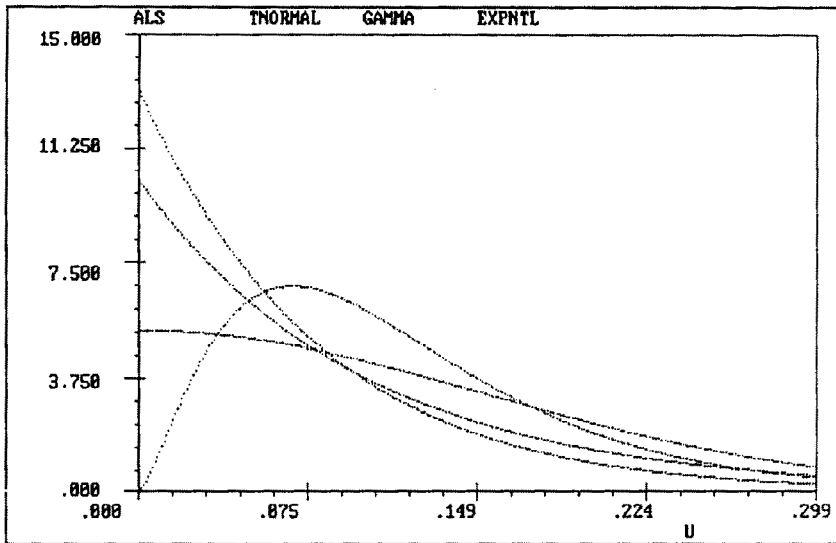


Fig. 1. Efficiency distributions.

7. Conclusions

The gamma model appears to offer a promising alternative to the half-normal and exponential models for the stochastic frontier. For our data, the half-normal, truncated normal, and exponential models give essentially the same distribution. But, the likelihood ratio and LM tests strongly reject the restriction of the exponential model. This suggests that the restriction in both the half-normal and exponential models may have a large influence on the pattern of the estimated inefficiency.

The estimated inefficiencies suggest that the restricted models (and Stevenson's extension) produce much larger values than the more general gamma distribution for most of the observations in the sample. But, for those data points for which the former two models suggest large inefficiency estimates, the gamma model gives largely the same conclusion. The upshot would seem to be that the single-parameter models are providing a more pessimistic impression than is warranted. Stevenson's extension of the ALS model bears some similarity to the gamma model in that its second parameter allows some greater flexibility in the shape of the distribution. Nonetheless, this model looks quite similar to the other two for these data, while the gamma model is qualitatively different. The marked difference suggested in fig. 1 is strongly suggestive.

Appendix

Table 3
Data used in the analysis of production costs.

	Cost function data					Estimated inefficiencies				
	Cost	Q	PL	PK	PF	TRN	ALS	EXP	GMA	OLS
Cal. Pac. Util.	0.2130	8.0	6869.47	64.945	18.0000	0.387	0.313	0.400	0.191	0.091
Montana Power	3.0427	869.0	8372.96	68.227	21.0670	0.028	0.036	0.027	0.002	0.008
Upper Pen. Pwr.	9.4059	1412.0	7960.90	40.692	41.5300	0.065	0.091	0.061	0.092	0.018
Mt. Carmel Pub.	0.7606	65.0	8791.89	41.243	28.5390	0.092	0.111	0.087	0.109	0.019
Bangor Hydro.	2.2587	295.0	8218.40	71.940	39.2000	0.039	0.051	0.038	0.075	0.011
Community P.S.	1.3422	183.0	5063.49	74.430	35.5100	0.043	0.054	0.041	0.079	0.011
Newport Elec.	0.6159	50.0	9204.24	90.470	32.0700	0.049	0.059	0.047	0.086	0.011
Maine Pub. Ser.	0.4887	14.0	5438.89	86.110	34.1500	0.486	0.392	0.504	0.206	0.225
N'western P.S.	1.1474	90.0	7189.67	79.101	21.5030	0.415	0.356	0.423	0.191	0.183
Pacific P & L	7.5492	2969.0	8183.34	80.657	9.0000	0.139	0.159	0.131	0.126	0.028
Central Kansas	2.0532	374.0	7884.94	82.485	26.3014	0.040	0.052	0.038	0.077	0.010
Ark. Mo. Power	0.6363	67.0	6696.50	58.258	25.4000	0.071	0.085	0.067	0.100	0.015
Lake Sup. Dist. Pr.	3.1504	378.0	7895.43	60.277	42.4883	0.055	0.073	0.052	0.087	0.014
Mo. Public Ser.	10.3136	1886.0	6833.93	67.680	25.6000	0.157	0.181	0.148	0.129	0.039
Mont. Dak. Utils.	5.8488	1025.0	7093.32	68.227	22.2793	0.189	0.205	0.180	0.139	0.047
United Gas. I.	4.5050	467.0	8410.34	76.300	26.8500	0.368	0.335	0.368	0.180	0.175
St. Joseph L & P	5.5971	938.0	7703.26	77.197	25.4000	0.128	0.152	0.120	0.121	0.029
Black Hills P & L	3.7255	643.0	7332.13	64.506	21.1619	0.162	0.181	0.153	0.132	0.037
Iowa Southern	6.0065	1328.0	6680.77	51.817	25.2319	0.063	0.085	0.059	0.092	0.016
Tucson Gas & E.	12.7830	2632.0	8469.84	70.295	30.9710	0.051	0.069	0.048	0.084	0.014
Madison Gas & E.	6.6214	856.0	8033.72	67.680	39.5812	0.091	0.118	0.023	0.106	0.022
Central La. Pwr.	9.6429	2689.0	6364.40	78.440	23.4889	0.045	0.059	0.042	0.080	0.012
Savannah E & P.	8.6852	1627.0	7912.40	79.220	31.6801	0.057	0.076	0.053	0.089	0.015
Otter Tail Pwr.	8.6372	1090.0	8507.92	39.127	33.9000	0.204	0.227	0.194	0.140	0.067
Cent. Maine Pwr.	11.2419	2258.0	6892.61	81.096	29.1000	0.073	0.097	0.068	0.098	0.018
Empire Dist. El.	6.6221	1500.0	7588.70	82.840	21.3951	0.080	0.102	0.074	0.102	0.018
El Paso Elec.	10.9665	2506.0	7412.36	57.240	27.4207	0.054	0.074	0.051	0.086	0.014

Table 3 (continued)

	Cost function data					Estimated inefficiencies				
	Cost	Q	PL	PK	PF	TRN	ALS	EXP	GMA	OLS
Interstate Pwr.	13.6270	2437.0	7846.06	65.565	30.6712	0.096	0.125	0.089	0.108	0.024
So. Ind. G.&E.	7.4395	1293.0	6821.09	78.225	21.4305	0.254	0.253	0.247	0.155	0.076
Iowa Pub. Ser.	9.7843	1627.0	8355.75	76.140	31.3374	0.090	0.117	0.084	0.106	0.022
P.S. Co. of N.H.	20.8671	3965.0	8403.59	74.480	33.1992	0.065	0.089	0.061	0.093	0.017
Ctrl. Hudson G.&E.	19.7092	2682.0	9484.85	81.750	50.4516	0.058	0.079	0.054	0.088	0.016
Iowa Ill. G.&E.	14.0431	2001.0	8611.53	74.025	30.6446	0.209	0.225	0.199	0.143	0.061
West Texas Util.	10.1902	2764.0	6338.00	66.232	21.1970	0.066	0.088	0.062	0.095	0.040
Iowa Elec. L.&Pwr.	16.1764	2487.0	8088.45	66.622	32.6186	0.152	0.180	0.143	0.127	0.025
So. Car. El.&Gas	42.2514	7320.0	5879.51	92.063	39.2104	0.097	0.126	0.091	0.109	0.021
Wisc. Pub. Ser.	22.5612	3571.0	7297.71	78.255	41.5951	0.079	0.106	0.074	0.101	0.025
No. Ind. Pub. Ser.	33.0175	6837.0	7310.15	69.795	28.4405	0.099	0.128	0.092	0.109	0.021
Rochester G.&E.	18.8963	2020.0	10806.20	79.570	46.9000	0.154	0.183	0.144	0.127	0.025
Iowa Pwr.&Light	13.2679	2445.0	8764.75	76.140	28.3648	0.093	0.120	0.086	0.107	0.042
Wisc. Pwr.&Light	21.5454	3981.0	8186.05	75.082	35.2049	0.064	0.087	0.059	0.092	0.022
Tampa Electric	35.5303	6770.0	7798.26	67.570	29.8250	0.115	0.146	0.107	0.115	0.017
Atl. City Elec.	29.8011	4187.0	7996.44	74.120	47.4257	0.086	0.116	0.080	0.103	0.030
United Ill. Co.	30.8773	5643.0	10182.49	61.040	27.8498	0.124	0.156	0.115	0.118	0.023
Ariz. Pub. Ser.	24.3565	6793.0	6336.88	70.295	18.5909	0.117	0.144	0.109	0.117	0.033
Kansas Pwr.&L.	17.4802	4148.0	7536.89	74.025	24.5837	0.071	0.095	0.066	0.097	0.028
Kansas Gas&El.	19.9008	5785.0	7969.55	71.910	22.2448	0.046	0.062	0.043	0.081	0.018
Louisiana P.&L.	28.7861	9660.0	6686.73	79.542	20.2630	0.044	0.058	0.042	0.080	0.012
Central Pwr.&L.	27.0832	7896.0	7119.96	74.513	20.1100	0.068	0.091	0.064	0.096	0.012
Miss. Power&L.	22.4421	5648.0	8954.12	78.440	25.9160	0.046	0.062	0.043	0.081	0.017
San Diego G.&E.	30.2067	5286.0	7084.10	73.325	38.3384	0.079	0.106	0.073	0.100	0.013
Pub. Ser. Okla.	30.1678	9602.0	7054.18	59.977	20.2010	0.052	0.071	0.049	0.085	0.021
Dallas Pwr.&L.	32.5840	7930.0	7119.01	48.997	22.8380	0.109	0.141	0.101	0.113	0.014
Florida Power	47.3864	9530.0	7624.57	83.880	31.5825	0.079	0.106	0.074	0.101	0.028
Ctrl. Ill. Pub. Ser.	24.2903	5316.0	9759.83	74.025	27.8380	0.058	0.079	0.541	0.089	0.020
Indy. Power&L.	31.2922	7484.0	8063.73	67.680	23.5267	0.087	0.114	0.081	0.105	0.015
Okla. Gas&Elec.	31.9884	10149.0	6437.92	73.140	18.5343	0.071	0.093	0.066	0.098	0.022

Texas Power & L.	41.9016	12954.0	6460.64	62.330	21.7550	0.053	0.072	0.050	0.086	0.017
Gulf States Util.	58.1154	17875.0	6288.41	73.395	20.6191	0.062	0.083	0.058	0.093	0.014
NY State El.& Gas	40.5281	6891.0	6769.55	74.120	35.9651	0.122	0.153	0.114	0.118	0.016
Kan. City P.& L.	37.0666	6754.0	10177.89	77.197	25.6208	0.165	0.190	0.155	0.131	0.032
Swestern P.S.	25.1686	6779.0	7826.93	74.200	20.2790	0.081	0.107	0.076	0.103	0.044
Texas Elec. Ser.	45.1827	12936.0	8320.06	65.760	22.0330	0.053	0.072	0.050	0.086	0.020
Long Is. Light	55.1764	10855.0	8061.96	71.490	31.7604	0.084	0.113	0.078	0.103	0.014
Illinois Power	41.1798	9275.0	8657.53	76.140	24.5804	0.093	0.121	0.086	0.107	0.022
Carolina P.& L.	80.3593	16311.0	7282.61	81.550	40.9692	0.042	0.058	0.040	0.078	0.023
Balt. Gas & El.	68.4800	12542.0	8142.84	80.385	35.7882	0.078	0.105	0.073	0.101	0.012
Potomac El. Pr.	97.3859	13846.0	7786.37	88.540	44.1571	0.129	0.160	0.121	0.121	0.021
Boston Edison	52.7634	9145.0	10373.50	81.750	35.8083	0.074	0.100	0.069	0.098	0.035
Ark. Power & Lt.	38.8472	10057.0	6035.95	81.578	25.8240	0.065	0.087	0.061	0.094	0.019
Pub. Ser. of Ind.	48.1125	11114.0	8413.86	69.975	22.5536	0.116	0.146	0.108	0.117	0.017
Wisc. Elec. Pwr.	79.0705	12706.0	9282.51	70.853	37.2477	0.109	0.141	0.101	0.113	0.209
Vir. Elec.&Pwr.	111.8863	23217.0	6873.73	83.880	33.3944	0.067	0.091	0.063	0.096	0.029
Penn. Pwr.& Lt.	77.8849	13702.0	7113.79	70.850	34.9616	0.120	0.151	0.112	0.117	0.018
Houston Lt.& Pr.	87.1015	27708.0	6378.23	63.600	20.3000	0.056	0.075	0.052	0.089	0.032
Duquesne Light	57.7267	10004.0	6472.86	76.300	28.0959	0.259	0.264	0.252	0.156	0.015
Consumers Pwr.	90.7168	17280.0	9191.47	72.967	36.8816	0.056	0.078	0.053	0.088	0.096
Pub. Ser. El.& G.	183.2315	27118.0	9914.36	78.480	41.7578	0.097	0.128	0.090	0.109	0.016
So. Cal. Edison	169.2354	38343.0	9117.16	65.992	31.5897	0.043	0.059	0.041	0.078	0.026
Niagara Mohawk	76.2528	11667.0	10436.32	80.660	46.0701	0.061	0.085	0.057	0.091	0.012
Phila. Elect.	134.2283	19445.0	9829.32	67.580	38.8027	0.145	0.177	0.135	0.125	0.017
Duke Power Co.	168.3777	34212.0	5683.83	80.385	40.5286	0.050	0.068	0.047	0.085	0.042
Pac. Gas & Elec.	125.3356	24001.0	8047.35	74.372	33.0932	0.083	0.111	0.078	0.103	0.014
Detroit Edison	191.5625	30958.0	9810.10	69.541	36.3076	0.103	0.135	0.096	0.111	0.022
Consol. Edison	240.5137	29613.0	9312.93	81.750	41.8972	0.225	0.241	0.216	0.147	0.228
Ctrl. Ver. Pub. Ser.	0.1304	4.0	6009.70	92.650	33.1990	0.050	0.053	0.050	0.093	0.080
Citizens Utils.	0.7293	60.0	7972.71	75.464	33.7070	0.059	0.073	0.057	0.093	0.009
Maui Electric	1.7705	153.0	10963.90	57.612	46.1600	0.049	0.064	0.046	0.083	0.013
Hilo Elec. Light	2.2367	198.0	7046.50	64.945	33.0000	0.161	0.178	0.153	0.132	0.012
Fitchburg G.& E.	2.5593	243.0	7000.81	81.750	33.4360	0.211	0.216	0.204	0.146	0.036
New Mex. Elec. Ser.	2.0358	617.0	9547.72	81.750	18.4900	0.024	0.030	0.023	0.001	0.023
Sierra Pac. Pwr.	7.6236	1340.0	6438.42	79.220	38.1483	0.049	0.065	0.046	0.083	0.007
Ctrl. Tel.& Util.	11.1091	1961.0	6964.81	82.485	28.2634	0.129	0.155	0.121	0.121	0.013

Table 3 (continued)

	Cost function data					Estimated inefficiencies				
	Cost	Q	PL	PK	PF	TRN	ALS	EXP	GMA	OLS
Pub. Ser. New Mex.	9.4674	2233.0	6717.18	59.957	22.6933	0.084	0.110	0.078	0.103	0.031
Nevada Power	13.8262	2582.0	8334.89	77.531	34.1811	0.055	0.075	0.052	0.087	0.020
Orange & Rocklin.	17.2895	2763.0	8058.91	80.660	45.6636	0.049	0.067	0.046	0.083	0.015
Kentucky Util.	16.1704	2863.0	7509.37	81.550	30.2374	0.115	0.143	0.107	0.116	0.014
Hawaiian Elec.	24.3975	3490.0	8477.23	75.420	38.0000	0.140	0.169	0.130	0.123	0.028
Toledo Edison	27.2443	4568.0	9619.76	76.140	35.6645	0.080	0.108	0.075	0.101	0.037
Colms. & So. Ohio	29.4870	5292.0	8176.33	76.140	22.8374	0.280	0.278	0.274	0.160	0.021
Dayton Pwr. & Lt.	35.1091	5699.0	7988.48	80.370	42.6351	0.070	0.095	0.065	0.096	0.107
Louisville G. & E.	25.4814	5702.0	6947.11	78.225	24.2596	0.114	0.142	0.106	0.116	0.019
Cinci. Gas & El.	42.4117	8650.0	7146.67	72.967	27.8693	0.120	0.150	0.112	0.118	0.028
Cleveland El. I.	72.0355	12724.0	8037.84	74.025	37.5174	0.082	0.110	0.076	0.102	0.030
Florida Pwr. & L.	146.9890	25147.0	9900.76	75.725	33.4742	0.104	0.135	0.097	0.112	0.022
New England El.	51.7415	10361.0	9578.63	68.016	28.1423	0.094	0.124	0.088	0.108	0.027
New Eng. G. & E. Ass.	21.5587	3886.0	9538.68	63.569	30.8894	0.090	0.120	0.083	0.105	0.024
East. Util. Ass.	14.2710	1901.0	9822.64	69.411	36.0000	0.146	0.175	0.137	0.125	0.023
Nrth. Sts. Pwr.	66.1032	11837.0	8709.43	75.379	31.3321	0.123	0.153	0.114	0.118	0.038
Union Elec. Co.	113.2555	22522.0	9500.78	76.732	25.0289	0.141	0.169	0.132	0.125	0.032
Allegheny Pr.	90.3718	21956.0	7954.47	83.338	22.9115	0.092	0.118	0.086	0.108	0.036
Southern Co.	240.4858	53918.0	6068.87	78.380	31.1954	0.059	0.079	0.056	0.091	0.022
Amer. Elec. Pr.	277.2965	72247.0	7419.92	56.301	25.0914	0.048	0.066	0.046	0.083	0.016
General Pub. U.	107.9776	18455.0	6690.23	76.300	32.9654	0.174	0.199	0.165	0.134	0.013
Common. Edison	269.7728	46870.0	9761.38	69.541	33.1999	0.091	0.120	0.085	0.107	0.050
Northeast Util.	79.6207	16508.0	9404.97	78.044	42.2086	0.032	0.043	0.030	0.000	0.024
Pub. Ser. Colo.	33.8814	7382.0	7512.72	72.362	25.9001	0.103	0.132	0.096	0.112	0.010
Minnesota P. & L.	14.9529	2325.0	8568.23	53.890	38.9161	0.078	0.107	0.072	0.099	0.026
Delmarva P. & L.	33.9733	5708.0	10024.20	78.102	42.1660	0.051	0.071	0.048	0.084	0.021
Ohio Edison Co.	78.7028	17132.0	8160.80	78.899	25.5378	0.105	0.134	0.098	0.113	0.014
Utah Power & Lt.	19.4391	4560.0	8558.37	76.464	23.7777	0.074	0.098	0.069	0.099	0.026
S'western El. Pr.	20.8836	6259.0	6697.02	69.764	21.2920	0.054	0.073	0.051	0.087	0.018
New Orleans P.S.	21.7792	6746.0	9419.27	49.778	20.1000	0.044	0.061	0.042	0.079	0.014

References

- Abramowitz, M. and I. Stegun, 1964, Handbook of mathematical functions (National Bureau of Standards, Government Printing Office, Washington, DC).
- Afriat, S., 1973, Efficiency estimation of production functions, *International Economic Review* 13, 568–598.
- Aguilar, R., 1988, Efficiency in production: Theory and an application on Kenyan smallholders, Ph.D. thesis (University of Goteborg, Goteborg).
- Aigner, D., C. Lovell, and P. Schmidt, 1977, Formulation and estimation of stochastic frontier production models, *Journal of Econometrics* 6, 21–37.
- Amemiya, T., 1973, Regression analysis when the dependent variable is truncated normal, *Econometrica* 41, 997–1016.
- Beckers, D. and C. Hammond, 1987, A tractable likelihood function for the normal-gamma stochastic frontier model, *Economics Letters* 24, 33–38.
- Berndt, E., B. Hall, R. Hall, and J. Hausman, 1974, Estimation and inference in nonlinear structural models, *Annals of Economic and Social Measurement* 3, 653–666.
- Fare, R., S. Grosskopf, and C. Lovell, 1983, The structure of technical efficiency, *Scandinavian Journal of Economics* 85, 181–190.
- Farrell, M. 1957, The measurement of productive efficiency, *Journal of the Royal Statistical Society A* 120, 253–281.
- Gjorlielsen, A., 1975, On estimating efficient production functions, Working paper no. A-35 (Chr. Michelsen Institute, Department of Humanities and Social Sciences, Bergen).
- Greene, W., 1980, Maximum likelihood estimation of econometric frontier functions, *Journal of Econometrics* 13, 27–56.
- Jondrow, J., C. Lovell, I. Materov, and P. Schmidt, 1982, On the estimation of technical inefficiency in the stochastic frontier production function model, *Journal of Econometrics* 19, 233–238.
- Kennedy, W. and J. Gentle, 1980, Statistical computing (Marcel-Dekker, New York, NY).
- Kopp, R. and W. Diewert, 1982, The decomposition of frontier cost function deviations into measures of technical and allocative efficiency, *Journal of Econometrics* 19, 319–332.
- Manski, C. and S. Lerman, 1977, The estimation of choice probabilities from choice based samples, in: C. Manski and D. McFadden, eds., *Structural analysis of discrete data with econometric applications* (MIT Press, Cambridge, MA).
- McFadden, D., 1989, A method of simulated moments for estimation of multinomial probits without numerical integration, *Econometrica* 57, 1195–1225.
- Meeusen, W. and J. van den Broeck, 1977, Efficiency estimation from Cobb–Douglas production functions with composed error, *International Economic Review* 18, 435–444.
- Newey, W., 1984, A method of moments interpretation of sequential estimators, *Economics Letters* 14, 201–206.
- Stevenson, R., 1980, Likelihood functions for generalized stochastic frontier estimation, *Journal of Econometrics* 13, 58–66.